

Punam Pukale

Software Engineer | FinTech & AI Platforms | Healthcare & Insurance | Agentic AI
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SUMMARY

Applied AI Engineer with 6+ years of experience delivering full-stack AI solutions across fintech, healthcare, insurance, and e-commerce. Currently designing React/TypeScript UIs and Python/Node.js APIs for multi-agent RAG workflows at Morgan Stanley, reducing compliance research time by 60%. Proven ability to own end-to-end AI experiences from complex client-side state management to secure microservice infrastructure and winner of the 2025 MIT Global AI Hackathon.

WORK EXPERIENCE

Morgan Stanley

Jan 2025 - Present

Software Engineer, GenAI & Agentic AI Platform

- Identified a critical gap in how compliance and risk teams consumed regulatory filings, with analysts spending hours per document in manual review. Architected an agentic RAG platform from scratch using Python, LangChain, LangGraph, and Weaviate with multi-hop retrieval and self-critique agents, backed by Claude Sonnet 4.5 and GPT-5 AI coding models, cutting analyst research time by 60% and reducing hallucinated citations by 35%.
- Integrated Model Context Protocol (MCP) to standardize agent access to internal tools and data sources, including document stores, compliance databases, and filing metadata APIs, replacing bespoke per-integration connectors and reducing new integration time from days to under 2 hours.
- Paired the backend platform with a full-stack analyst review interface in React, TypeScript, and TailwindCSS that surfaced LLM quality metrics and real-time feedback loops using sophisticated state management. This product decision proved critical to adoption and lifted response relevance scores by 28% within six weeks.
- Designed the multi-agent orchestration architecture, choosing to sequence document retrieval, embedding generation, and async LLM inference in FastAPI and Express rather than a monolithic pipeline, enabling fully autonomous financial summarization without analyst handoff and facilitating advanced financial analytics workflows.
- Hardened the API gateway with OAuth 2.0, JWT validation, rate-limiting, and payload schema enforcement before any external teams onboarded. This proactive security posture sustained zero authentication incidents across 18K+ monthly requests.
- Designed horizontal pod autoscaling on AWS EKS with Helm-managed Kubernetes ahead of anticipated growth. When a 3x traffic surge arrived from new financial product lines, the platform absorbed it without incident and held p99 latency under 420ms.
- Invested in OpenTelemetry distributed tracing and structured CloudWatch log pipelines before failures occurred. The upfront engineering cost paid off by cutting mean time to detect LLM inference failures by 50% and enabling sub-minute root-cause isolation during live incidents.
- Personally coached 2 junior engineers on AI architecture patterns, MCP tool integration, Weaviate vector index tuning, and production incident runbooks, compressing their onboarding by 3 weeks and halving P2 escalations to senior engineers.

Apexon

Jul 2022 - Apr 2023

Senior Software Engineer, Healthcare

- Scoped and led delivery of a HIPAA-compliant clinical trial oversight platform covering 15+ study sites. Proposed replacing ad-hoc CSV reporting with live enrollment tracking, site compliance scoring, and protocol deviation alerts using Redux Toolkit, cutting reporting turnaround by 35% while implementing business intelligence tools for real-time insights.
- Replaced a manual CSV export workflow for 200K+ clinical records with interactive D3.js SVG visualisations giving clinical leadership same-day visibility into adverse events and safety signals.
- Diagnosed a data pipeline throughput bottleneck and decided to migrate high-frequency read endpoints from REST to gRPC, a significant architectural trade-off, paired with React Query prefetch strategies. This achieved a 40% throughput improvement and sustained sub-300ms response times at peak quarterly reporting.
- Designed HIPAA data governance from the ground up, including field-level PII masking, session-scoped access tokens, and tamper-evident audit trails across 200+ investigators. Passed all regulatory access audits with zero findings.

Hexaware Technologies

Oct 2020 - Jun 2022

Product Developer, Insurance

- Built the LV.com insurance policy servicing platform using TypeScript microservices and a React component library managing quotes, endorsements, and full policy lifecycle across 50K+ active records, improving agent servicing efficiency by 32%.
- Eliminated 28% of manual underwriting review effort by engineering server-side Node.js eligibility rule engines paired with controlled React form components that auto-surface ineligibility reasons during high-volume open enrollment cycles.
- Powered sub-second retrieval across 1M+ insurance policy records by integrating Elasticsearch with a GraphQL search gateway, replacing a legacy SQL LIKE query approach that was causing consistent agent workflow timeouts.
- Architected an event-driven policy synchronization layer using RabbitMQ message queues to propagate 20K+ daily updates across customer, billing, and policy microservices in real time, eliminating reconciliation defects that previously required overnight batch correction.

- Profiled and resolved 34% of UI latency on high-traffic agent servicing views through Chrome DevTools flame graph analysis, targeted PostgreSQL composite index additions, and Brotli compression on API response payloads.

Primal Infosys

Jan 2019 - Sep 2020

Software Developer, E-Commerce

- Engineered an e-commerce customer support and order management platform in JavaScript ES6 and React.js with Express REST services, handling 25K+ monthly customer interactions and cutting average support resolution time by 30% through rule-based ticket routing.
- Processed 8K+ daily order and account management transactions by building Express and MySQL backend services with prepared statement query optimization, read replica routing, and connection pool tuning for consistent sub-200ms API responses.
- Authored SLA-aware ticket prioritization logic using Bull queue workers that scored and routed support cases by order value, issue category, and customer tier, reducing backlog accumulation by 25% and eliminating the manual triage queue entirely.
- Trimmed 36% off repeat-visit dashboard load times by introducing Redis key-value caching on high-frequency product and order queries, normalizing Redux store shape to minimize selector recomputation, and enabling HTTP response compression across all API routes.

ACHIEVEMENTS

- First Prize, 2025 Global AI Hackathon at MIT, 1,300+ participants across 86 countries
- Published IEEE Paper, AllIoT 2025 Conference

PROJECT

Autonomous Software Engineering Agent - End-to-End AI Software Delivery Platform

github.com/PUNAMRAMPUKALE/autonomous-software-agent

- Designed a 14-agent autonomous pipeline leveraging agentic development workflows to carry a Jira story from backlog selection through code generation, automated review, testing, PR creation, and Jira closure with zero manual intervention
- Built a repository intelligence layer that clones and parses target repos, detects language and framework, indexes source symbols, and builds a dependency graph to retrieve minimal, relevant context for LLM prompting on large codebases.
- Implemented a provider-agnostic LLM API layer (OpenAI, Claude, Gemini, Ollama) paired with a validation engine that converts AI-generated output into structured file operations before applying changes, automating the full delivery loop including testing, Git push, PR creation, Jira updates, and Slack notifications
- Tech Stack: Python 3.11, OpenAI API, Anthropic Claude API, Gemini, Ollama, PyGithub, Jira REST API, GitPython, Slack API, python-dotenv.

EDUCATION

University of the Pacific

Aug 2023 - May 2025

Master of Science, Computer Science

Stockton, CA

SKILLS & COMPETENCIES

Languages: TypeScript, JavaScript (ES2022+), Python, Java, SQL

Frontend: React, Redux Toolkit, React Query, TailwindCSS, D3.js, react-window, Storybook, Webpack, Vite, Web Performance Optimization, WCAG Accessibility, State Management

Backend & APIs: Node.js, Express, FastAPI, gRPC, REST API Design, tRPC, WebSockets, Kafka, RabbitMQ, Bull Queue, Redis, PostgreSQL, MySQL, MongoDB, DynamoDB

Agentic AI & LLMs: LangChain, LangGraph, Model Context Protocol (MCP), OpenAI API (function calling, tool use), Agentic RAG, RAG Pipeline Design, Vector Search (Weaviate, Pinecone), HuggingFace Transformers, Prompt Engineering, Multi-Agent Orchestration, Retrieval Augmented Generation, AI Coding Models, Agentic Development Workflows

Distributed Systems & Architecture: Microservices Architecture, Event-Driven Design, System Design (HLD/LLD), API Gateway Patterns, CQRS, Saga Pattern, Rate Limiting, Circuit Breakers, High-Availability Design

Containerization & Cloud: Docker, Kubernetes (AWS EKS, GKE, Azure AKS), Helm, Terraform, ArgoCD, AWS (EC2, S3, Lambda, RDS, SQS, IAM, CloudWatch), GCP Firebase, CI/CD (GitHub Actions, Jenkins), GitOps, Blue-Green and Canary Deployments

Observability & Reliability: OpenTelemetry, Prometheus, Grafana, Jaeger, CloudWatch, Structured Logging, Distributed Tracing, Incident Response Runbooks, SLO/SLA Design, MTTR/MTTD Optimization

ML & MLOps: PyTorch, TensorFlow, scikit-learn, MLflow, Model Versioning, Feature Engineering, Real-Time and Batch Inference Pipelines, Airflow

Domain & Compliance: FinTech AI Systems, Healthcare Clinical Trials (HIPAA), Insurance Policy Platforms, E-Commerce, OAuth 2.0, JWT, OpenAPI Contracts, Agile/Scrum, Technical Spec Writing, Code Review Leadership, Business Intelligence Tools, Data Visualization, Financial Analytics